

Algebra & Equations

ANSWER KEY



Solving linear equations

1 Solve $3x - 7 = 11$.

$$3x = 18, \text{ so } x = 6.$$

2 Solve $5(x - 2) = 3x + 4$.

$$5x - 10 = 3x + 4 \Rightarrow 2x = 14 \Rightarrow x = 7.$$

3 Solve $\frac{3x - 2}{4} = x - 1$.

$$3x - 2 = 4(x - 1) = 4x - 4 \Rightarrow -x = -2 \Rightarrow x = 2.$$

4 Solve $2(x + 3) - 5 = 3(x - 1)$.

$$2x + 1 = 3x - 3 \Rightarrow -x = -4 \Rightarrow x = 4.$$

Expanding and simplifying expressions

5 Expand and simplify $3(2x - 1) - 2(x - 4)$.

$$6x - 3 - 2x + 8 = 4x + 5.$$

6 Expand $(x + 5)(x - 2)$.

$$x^2 - 2x + 5x - 10 = x^2 + 3x - 10.$$

7 Expand $(2x - 3)^2$.

$$(2x - 3)^2 = 4x^2 - 12x + 9.$$

8 Simplify $\frac{6x^2 - 9x}{3x}$ (for $x \neq 0$).

$$\frac{3x(2x - 3)}{3x} = 2x - 3.$$

Factoring quadratic expressions

9 Factor $x^2 + 7x + 12$.

$$(x + 3)(x + 4).$$

10 Factor $x^2 - 2x - 15$.

$$(x - 5)(x + 3).$$

11 Factor $2x^2 + 5x - 3$.

$$(2x - 1)(x + 3).$$

12 Factor $x^2 - 49$ using the difference of squares.

$$(x - 7)(x + 7).$$

Solving quadratic equations by factoring

13 Solve $x^2 - 5x + 6 = 0$.

$$(x - 2)(x - 3) = 0, \text{ so } x = 2 \text{ or } x = 3.$$

14 Solve $x^2 + 2x - 8 = 0$.

$$(x + 4)(x - 2) = 0, \text{ so } x = -4 \text{ or } x = 2.$$

15 Solve $2x^2 - 5x - 3 = 0$.

$$(2x + 1)(x - 3) = 0, \text{ so } x = -\frac{1}{2} \text{ or } x = 3.$$

16 Solve $x^2 - 9x + 20 = 0$.

$$(x - 4)(x - 5) = 0, \text{ so } x = 4 \text{ or } x = 5.$$

The quadratic formula and the discriminant

17 Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

$$a = 1, b = 3, c = 1. x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{-3 \pm \sqrt{5}}{2}.$$

18 Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

$$\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24. \text{ Since } \Delta < 0, \text{ there are no real roots.}$$

19 Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.

$$\Delta = (-6)^2 - 4(1)(9) = 36 - 36 = 0. \text{ Since } \Delta = 0, \text{ there is one repeated real root } (x = 3).$$

20 Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.

$$\Delta = (-2)^2 - 4(3)(-1) = 4 + 12 = 16. x = \frac{2 \pm 4}{6}, \text{ so } x = 1 \text{ or } x = -\frac{1}{3}.$$

Absolute value equations

- 21** Solve $|x - 4| = 9$.

$$x - 4 = 9 \Rightarrow x = 13, \text{ or } x - 4 = -9 \Rightarrow x = -5.$$

- 22** Solve $|2x + 3| = 7$.

$$2x + 3 = 7 \Rightarrow x = 2, \text{ or } 2x + 3 = -7 \Rightarrow x = -5.$$

- 23** Solve $3|x - 1| = 12$.

$$|x - 1| = 4, \text{ so } x - 1 = 4 \Rightarrow x = 5, \text{ or } x - 1 = -4 \Rightarrow x = -3.$$

Linear inequalities

- 24** Solve $2x - 5 < 9$ and graph the solution set on a number line.

$$2x < 14 \Rightarrow x < 7.$$

- 25** Solve $-3x + 4 \geq -8$.

$$-3x \geq -12. \text{ Dividing by } -3 \text{ reverses the inequality: } x \leq 4.$$

- 26** Solve the compound inequality $5 < 2x + 1 \leq 11$.

$$\text{Subtract 1: } 4 < 2x \leq 10. \text{ Divide by 2: } 2 < x \leq 5.$$

Systems of linear equations

- 27** Solve the system $x + y = 10$, $x - y = 2$.

$$\text{Adding the equations: } 2x = 12 \Rightarrow x = 6. \text{ Then } y = 10 - 6 = 4.$$

- 28** Solve the system $3x + 2y = 16$, $x - y = 2$.

$$\text{From the second equation, } x = 2 + y. \text{ Substituting: } 3(2 + y) + 2y = 16 \Rightarrow 6 + 5y = 16 \Rightarrow y = 2. \text{ Then } x = 4.$$

- 29** This question has two parts. The sum of two numbers is 15 and their difference is 3.

a Write a system of two equations for the numbers x and y . $x + y = 15$ and $x - y = 3$.

b Solve the system to find both numbers. Adding the equations: $2x = 18 \Rightarrow x = 9$. Then $y = 15 - 9 = 6$.

Rational equations

30 Solve $\frac{x}{x-2} = 3$.

$$x = 3(x-2) = 3x - 6 \Rightarrow -2x = -6 \Rightarrow x = 3.$$

31 Solve $\frac{2}{x+1} = \frac{1}{3}$.

Cross-multiplying: $6 = x + 1 \Rightarrow x = 5$.

32 Solve $\frac{x+1}{x-1} = 2$.

$$x + 1 = 2(x - 1) = 2x - 2 \Rightarrow -x = -3 \Rightarrow x = 3.$$

Solving for a variable in a formula

33 The area of a triangle is $A = \frac{1}{2}bh$. Solve for h .

$$h = \frac{2A}{b}.$$

34 The circumference of a circle is $C = 2\pi r$. Solve for r .

$$r = \frac{C}{2\pi}.$$

35 The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t .

$$t = \frac{v - v_0}{a}.$$

Linear-quadratic systems

36 Solve the system $y = x^2 - 1$, $y = 2x + 2$.

Setting equal: $x^2 - 1 = 2x + 2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. So $x = 3, y = 8$ or $x = -1, y = 0$.

37 Solve the system $y = x^2$, $y = 3x - 2$.

Setting equal: $x^2 = 3x - 2 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x - 1)(x - 2) = 0$. So $x = 1, y = 1$ or $x = 2, y = 4$.

Word problems: setting up and solving equations

- 38** This question has two parts. A rectangular garden has a length that is 4 meters more than its width w . Its area is 96 square meters.
- a** Write an equation relating w to the area. $w(w + 4) = 96$, i.e. $w^2 + 4w - 96 = 0$.
- b** Solve for w , and state the garden's length. Factoring: $(w - 8)(w + 12) = 0$. Since width is positive, $w = 8$ meters, so the length is $w + 4 = 12$ meters.
- 39** This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.
- a** Write a system of two equations for f and r . $f + 5r = 65$ and $f + 9r = 93$.
- b** Solve the system to find the fixed fee and the rate per gigabyte. Subtracting the first equation from the second: $4r = 28 \Rightarrow r = 7$. Then $f = 65 - 5(7) = 30$. Fixed fee: 30 dollars; rate: 7 dollars per GB.
-

MCQ — Solving linear equations

- 40** Solve $4x - 9 = 15$.

- a.** $x = 6$ ✓
- b.** $x = -6$
- c.** $x = 24$
- d.** $x = 1.5$

$$4x = 24 \Rightarrow x = 6.$$

- 41** Solve $6(x - 3) = 2x + 10$.

- a.** $x = -7$
- b.** $x = 4$
- c.** $x = 7$ ✓
- d.** $x = 1$

$$6x - 18 = 2x + 10 \Rightarrow 4x = 28 \Rightarrow x = 7.$$

42 Solve $\frac{2x-1}{3} = x-2$.

a. $x = 5$ ✓

b. $x = 1$

c. $x = 7$

d. $x = -5$

$$2x - 1 = 3x - 6 \Rightarrow x = 5.$$

43 Solve $3(x+4) - 7 = 2(x-1)$.

a. $x = -7$ ✓

b. $x = -3$

c. $x = 3$

d. $x = 7$

$$3x + 5 = 2x - 2 \Rightarrow x = -7.$$

44 Solve $5x + 3 = 2x - 9$.

a. $x = -2$

b. $x = 2$

c. $x = 4$

d. $x = -4$ ✓

$$3x = -12 \Rightarrow x = -4.$$

MCQ — Expanding and simplifying expressions

45 Expand and simplify $4(3x - 2) - 3(x - 5)$.

a. $15x - 23$

b. $9x + 23$

c. $9x + 7$ ✓

d. $9x - 23$

$$12x - 8 - 3x + 15 = 9x + 7.$$

46 Expand $(x + 7)(x - 3)$.

a. $x^2 + 10x - 21$

b. $x^2 - 4x - 21$

c. $x^2 + 4x - 21$ ✓

d. $x^2 + 4x + 21$

$$(x + 7)(x - 3) = x^2 + 4x - 21.$$

47 Expand $(3x - 1)^2$.

a. $3x^2 - 6x + 1$

b. $9x^2 - 1$

c. $9x^2 - 6x + 1$ ✓

d. $9x^2 - 6x - 1$

$$(3x - 1)^2 = 9x^2 - 6x + 1.$$

48 Simplify $\frac{8x^2 - 12x}{4x}$ (for $x \neq 0$).

a. $2x - 12$

b. $8x - 3$

c. $2x - 3$ ✓

d. $2x^2 - 3$

$$\frac{4x(2x - 3)}{4x} = 2x - 3.$$

MCQ — Factoring quadratic expressions

49 Factor $x^2 + 9x + 14$.

a. $(x + 2)(x + 7)$ ✓

b. $(x - 2)(x - 7)$

c. $(x + 1)(x + 14)$

d. $(x + 14)(x - 1)$

Need two numbers multiplying to 14, adding to 9: 2, 7.

50 Factor $x^2 - 3x - 18$.

a. $(x - 9)(x + 2)$

b. $(x - 2)(x + 9)$

c. $(x - 6)(x + 3)$ ✓

d. $(x + 6)(x - 3)$

Need two numbers multiplying to -18 , adding to -3 : $-6, 3$.

51 Factor $3x^2 + 7x + 2$.

a. $(3x + 1)(x + 2)$ ✓

b. $(3x - 1)(x - 2)$

c. $(3x + 2)(x + 1)$

d. $(x + 1)(3x + 2)$

$$(3x + 1)(x + 2) = 3x^2 + 6x + x + 2 = 3x^2 + 7x + 2.$$

52 Factor $x^2 - 81$ using the difference of squares.

a. $(x - 9)^2$

b. $(x - 9)(x + 9)$ ✓

c. $(x + 9)^2$

d. $(x - 81)(x + 1)$

$$x^2 - 9^2 = (x - 9)(x + 9).$$

53 Factor $2x^2 - x - 6$.

a. $(2x + 3)(x - 2)$ ✓

b. $(2x - 3)(x + 2)$

c. $(x + 3)(2x - 2)$

d. $(2x + 1)(x - 6)$

$$(2x + 3)(x - 2) = 2x^2 - 4x + 3x - 6 = 2x^2 - x - 6.$$

MCQ — Solving quadratic equations by factoring

54 Solve $x^2 - 7x + 10 = 0$.

a. $x = 1$ or $x = 10$

b. $x = 7$ or $x = 10$

c. $x = -2$ or $x = -5$

d. $x = 2$ or $x = 5$ ✓

$$(x - 2)(x - 5) = 0 \Rightarrow x = 2 \text{ or } x = 5.$$

55 Solve $x^2 + 3x - 10 = 0$.

- a. $x = -3$ or $x = 10$
- b. $x = -10$ or $x = 1$
- c. $x = 5$ or $x = -2$
- d. $x = -5$ or $x = 2$ ✓

$$(x + 5)(x - 2) = 0 \Rightarrow x = -5 \text{ or } x = 2.$$

56 Solve $3x^2 - 11x - 4 = 0$.

- a. $x = -\frac{1}{3}$ or $x = 4$ ✓
- b. $x = \frac{1}{3}$ or $x = -4$
- c. $x = 1$ or $x = 4$
- d. $x = -\frac{1}{3}$ or $x = -4$

$$(3x + 1)(x - 4) = 0 \Rightarrow x = -\frac{1}{3} \text{ or } x = 4.$$

57 Solve $x^2 - 11x + 24 = 0$.

- a. $x = 1$ or $x = 24$
- b. $x = 4$ or $x = 6$
- c. $x = -3$ or $x = -8$
- d. $x = 3$ or $x = 8$ ✓

$$(x - 3)(x - 8) = 0 \Rightarrow x = 3 \text{ or } x = 8.$$

MCQ — The quadratic formula and the discriminant

58 Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$ ✓

Discriminant $= 25 - 8 = 17$; $x = \frac{-5 \pm \sqrt{17}}{2}$.

59 Find the discriminant of $3x^2 - 6x + 5 = 0$ and state how many real roots it has.

a. 96; two real roots

b. 24; two real roots

c. -24; no real roots ✓

d. -24; one real root

$(-6)^2 - 4(3)(5) = 36 - 60 = -24 < 0$: no real roots.

60 Find the discriminant of $x^2 - 8x + 16 = 0$ and state how many real roots it has.

a. 0; one real root ✓

b. 64; two real roots

c. 0; two real roots

d. -64; no real roots

$(-8)^2 - 4(1)(16) = 64 - 64 = 0$: one repeated real root.

61 Use the quadratic formula to solve $2x^2 + 3x - 2 = 0$.

a. $x = 1$ or $x = -2$

b. $x = -0.5$ or $x = 2$

c. $x = 0.5$ or $x = -2$ ✓

d. $x = 0.5$ or $x = 2$

Discriminant $= 9 + 16 = 25$; $x = \frac{-3 \pm 5}{4}$: $x = 0.5$ or $x = -2$.

MCQ — Absolute value equations

62 Solve $|x - 6| = 11$.

- a. $x = -17$ or $x = 5$
- b. $x = 5$ or $x = -17$
- c. $x = 17$ or $x = -5$ ✓
- d. $x = 17$ or $x = 5$

$$x - 6 = 11 \text{ or } x - 6 = -11: x = 17 \text{ or } x = -5.$$

63 Solve $|3x + 2| = 13$.

- a. $x = \frac{11}{3}$ or $x = 5$
- b. $x = 5$ or $x = -5$
- c. $x = -\frac{11}{3}$ or $x = 5$
- d. $x = \frac{11}{3}$ or $x = -5$ ✓

$$3x + 2 = 13 \Rightarrow x = \frac{11}{3}; 3x + 2 = -13 \Rightarrow x = -5.$$

64 Solve $4|x - 2| = 20$.

- a. $x = 7$ or $x = -3$ ✓
- b. $x = -7$ or $x = 3$
- c. $x = 5$ or $x = -5$
- d. $x = 7$ or $x = 3$

$$|x - 2| = 5 \Rightarrow x - 2 = 5 \text{ or } x - 2 = -5: x = 7 \text{ or } x = -3.$$

65 Solve $|2x - 5| = 9$.

a. $x = 7$ or $x = 2$

b. $x = -7$ or $x = 2$

c. $x = 2$ or $x = -7$

d. $x = 7$ or $x = -2$ ✓

$$2x - 5 = 9 \Rightarrow x = 7; 2x - 5 = -9 \Rightarrow x = -2.$$

66 The graph shows $y = |x - 3|$. At what x -value does the graph reach its minimum?

a. $x = 1$

b. $x = 3$ ✓

c. $x = -3$

d. $x = 0$

The vertex of the V-shape (the minimum) is at $x = 3$.

MCQ — Linear inequalities

67 Solve $3x - 7 < 14$, and identify the graph of its solution set.

a. $x < 7$ ✓

b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

$$3x < 21 \Rightarrow x < 7.$$

68 Solve $-4x + 6 \geq -10$, and identify the graph of its solution set.

a. $x > 4$

b. $x < 4$

c. $x \leq 4$ ✓

d. $x \geq 4$

$-4x \geq -16$; dividing by -4 flips the sign: $x \leq 4$.

69 Solve the compound inequality $7 < 3x + 1 \leq 19$.

a. $2 < x \leq 6$ ✓

b. $2 < x < 6$

c. $2 \leq x < 6$

d. $6 < x \leq 2$

$6 < 3x \leq 18 \Rightarrow 2 < x \leq 6$.

70 Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$ ✓

$2x + 5 > -3 \Rightarrow x > -4$; $x - 1 < 6 \Rightarrow x < 7$. Together: $-4 < x < 7$.

MCQ — Systems of linear equations

71 Solve the system $x + y = 14$, $x - y = 4$.

- a. $x = 5, y = 9$
- b. $x = 10, y = 4$
- c. $x = 9, y = 5$ ✓
- d. $x = 9, y = 4$

Adding: $2x = 18 \Rightarrow x = 9$; then $y = 5$.

72 Solve the system $4x + 3y = 26$, $x - y = 3$.

- a. $x = 5, y = 2$ ✓
- b. $x = 2, y = 5$
- c. $x = 8, y = 5$
- d. $x = 5, y = 3$

$x = y + 3 \Rightarrow 4(y + 3) + 3y = 26 \Rightarrow 7y = 14 \Rightarrow y = 2$; $x = 5$.

73 The sum of two numbers is 22 and their difference is 6. Find the two numbers.

- a. 11 and 11
- b. 14 and 6
- c. 16 and 6
- d. 14 and 8 ✓

$x + y = 22$, $x - y = 6 \Rightarrow x = 14$, $y = 8$.

74 Solve the system $2x - y = 7$, $3x + 2y = 21$.

- a. $x = 5, y = -3$
- b. $x = 5, y = 3$ ✓
- c. $x = 3, y = 5$
- d. $x = 7, y = 7$

$y = 2x - 7 \Rightarrow 3x + 2(2x - 7) = 21 \Rightarrow 7x = 35 \Rightarrow x = 5$; $y = 3$.

MCQ — Rational equations

75 Solve $\frac{x}{x-3} = 4$.

a. $x = 12$

b. $x = -4$

c. $x = 4$ ✓

d. $x = 3$

$$x = 4(x - 3) = 4x - 12 \Rightarrow -3x = -12 \Rightarrow x = 4.$$

76 Solve $\frac{3}{x+2} = \frac{1}{4}$.

a. $x = 6$

b. $x = 14$

c. $x = 10$ ✓

d. $x = -10$

$$\text{Cross-multiply: } 12 = x + 2 \Rightarrow x = 10.$$

77 Solve $\frac{x+2}{x-2} = 3$.

a. $x = 2$

b. $x = 8$

c. $x = 4$ ✓

d. $x = -4$

$$x + 2 = 3(x - 2) = 3x - 6 \Rightarrow 8 = 2x \Rightarrow x = 4.$$

78 Solve $\frac{4}{x-1} = \frac{2}{x+2}$.

a. $x = -5$ ✓

b. $x = 5$

c. $x = -1$

d. $x = 2$

Cross-multiply: $4(x+2) = 2(x-1) \Rightarrow 4x+8 = 2x-2 \Rightarrow x = -5$.

79 Solve $\frac{6}{x} = \frac{2}{x-4}$.

a. $x = 12$

b. $x = 4$

c. $x = -6$

d. $x = 6$ ✓

Cross-multiply: $6(x-4) = 2x \Rightarrow 6x-24 = 2x \Rightarrow x = 6$.

MCQ — Solving for a variable in a formula

80 The perimeter of a rectangle is $P = 2l + 2w$. Solve for l .

a. $l = \frac{P}{2} - w$

b. $l = \frac{P-2w}{2}$ ✓

c. $l = \frac{P+2w}{2}$

d. $l = P - 2w$

$P - 2w = 2l \Rightarrow l = \frac{P-2w}{2}$ (equivalently $\frac{P}{2} - w$).

81 The formula for simple interest is $I = Prt$. Solve for r .

a. $r = \frac{I}{P}$

b. $r = \frac{I}{Pt}$ ✓

c. $r = \frac{Pt}{I}$

d. $r = I - Pt$

Dividing both sides by Pt : $r = \frac{I}{Pt}$.

82 The formula $F = \frac{9}{5}C + 32$ converts Celsius to Fahrenheit. Solve for C .

a. $C = \frac{5(F - 32)}{9}$ ✓

b. $C = \frac{5F - 32}{9}$

c. $C = \frac{5}{9}F + 32$

d. $C = \frac{9(F - 32)}{5}$

$$F - 32 = \frac{9}{5}C \Rightarrow C = \frac{5(F - 32)}{9}.$$

83 The volume of a cylinder is $V = \pi r^2 h$. Solve for h .

a. $h = \frac{V}{\pi r^2}$ ✓

b. $h = V - \pi r^2$

c. $h = \frac{V}{\pi r}$

d. $h = \frac{\pi r^2}{V}$

Dividing both sides by πr^2 : $h = \frac{V}{\pi r^2}$.

MCQ — Linear-quadratic systems

84 Solve the system $y = x^2 - 4$, $y = 3x$.

- a. $(-4, -12)$ and $(1, 3)$
- b. $(-1, -3)$ and $(1, 3)$
- c. $(4, 12)$ and $(1, 3)$
- d. $(4, 12)$ and $(-1, -3)$ ✓

$$x^2 - 4 = 3x \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0: x = 4, y = 12 \text{ or } x = -1, y = -3.$$

85 Solve the system $y = x^2$, $y = 2x + 3$.

- a. $(-3, 9)$ and $(1, 1)$
- b. $(3, 9)$ and $(1, 1)$
- c. $(-1, 1)$ and $(1, 1)$
- d. $(3, 9)$ and $(-1, 1)$ ✓

$$x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0: x = 3, y = 9 \text{ or } x = -1, y = 1.$$

86 The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?

- a. $(4, 12)$ and $(1, 3)$
- b. $(-4, -12)$ and $(1, 3)$
- c. $(-1, -3)$ and $(1, 3)$
- d. $(4, 12)$ and $(-1, -3)$ ✓

The curve and line cross at $(-1, -3)$ and $(4, 12)$.

87 The graph shows the system $y = x^2$, $y = 2x + 3$. What are the points of intersection?

- a. $(3, 9)$ and $(1, 1)$
- b. $(-1, 1)$ and $(1, 1)$
- c. $(-3, 9)$ and $(1, 1)$
- d. $(3, 9)$ and $(-1, 1)$ ✓

The curve and line cross at $(-1, 1)$ and $(3, 9)$.

MCQ — Word problems: setting up and solving equations

- 88** A rectangular garden has a length that is 5 meters more than its width, and its area is 84 square meters. Find the width.

a. 12

b. 7 ✓

c. 9

d. 14

$$w(w + 5) = 84 \Rightarrow (w + 12)(w - 7) = 0; \text{ the positive root is } w = 7.$$

- 89** A mobile-data plan charges a fixed monthly fee of 40 riyals plus 2 riyals per GB used. If a user's bill was 68 riyals, how many GB did they use?

a. 34

b. 14 ✓

c. 54

d. 28

$$40 + 2g = 68 \Rightarrow 2g = 28 \Rightarrow g = 14.$$

Multiple choice — easy

- 90** Solve $4(x - 1) = x + 11$.

a. -5

b. 3

c. 5 ✓

d. 7

$$4x - 4 = x + 11 \Rightarrow 3x = 15 \Rightarrow x = 5.$$

91 Solve $4(x + 9) = x + 36$.

a. 3

b. -2

c. 0 ✓

d. 2

$$4x + 36 = x + 36 \Rightarrow 3x = 0 \Rightarrow x = 0.$$

92 Solve $4(x - 1) = 2x - 8$.

a. 2

b. 0

c. -4

d. -2 ✓

$$4x - 4 = 2x - 8 \Rightarrow 2x = -4 \Rightarrow x = -2.$$

93 Solve $4(x + 7) = 3x + 25$.

a. 3

b. -1

c. -3 ✓

d. -5

$$4x + 28 = 3x + 25 \Rightarrow x = -3.$$

94 Solve $5(x + 9) = x + 89$.

a. -11

b. 11 ✓

c. 9

d. 13

$$5x + 45 = x + 89 \Rightarrow 4x = 44 \Rightarrow x = 11.$$

95 Solve $2(x + 7) = 3x + 23$.

a. -11

b. -7

c. -9 ✓

d. 9

$$2x + 14 = 3x + 23 \Rightarrow -x = 9 \Rightarrow x = -9.$$

96 Solve $5(x + 6) = x + 78$.

a. 10

b. 12 ✓

c. -12

d. 14

$$5x + 30 = x + 78 \Rightarrow 4x = 48 \Rightarrow x = 12.$$

97 Solve $4(x - 9) = x - 12$.

a. 10

b. 8 ✓

c. -8

d. 6

$$4x - 36 = x - 12 \Rightarrow 3x = 24 \Rightarrow x = 8.$$

98 Solve $5(x - 1) = x - 9$.

a. 2

b. -3

c. 1

d. -1 ✓

$$5x - 5 = x - 9 \Rightarrow 4x = -4 \Rightarrow x = -1.$$

99 Solve $5(x + 6) = x + 18$.

a. -1

b. -5

c. 3

d. -3 ✓

$$5x + 30 = x + 18 \Rightarrow 4x = -12 \Rightarrow x = -3.$$

100 Solve $5(x - 2) = 2x - 22$.

a. -2

b. -6

c. 4

d. -4 ✓

$$5x - 10 = 2x - 22 \Rightarrow 3x = -12 \Rightarrow x = -4.$$

101 Solve $2(x - 1) = x + 7$.

a. 9 ✓

b. -9

c. 11

d. 7

$$2x - 2 = x + 7 \Rightarrow x = 9.$$

102 Solve $2(x - 3) = 3x - 11$.

a. 3

b. 5 ✓

c. -5

d. 7

$$2x - 6 = 3x - 11 \Rightarrow -x = -5 \Rightarrow x = 5.$$

103 Solve $5(x + 9) = x + 57$.

a. 5

b. 1

c. 3 ✓

d. -3

$$5x + 45 = x + 57 \Rightarrow 4x = 12 \Rightarrow x = 3.$$

104 Solve $2(x + 7) = x + 10$.

a. 4

b. -2

c. -4 ✓

d. -6

$$2x + 14 = x + 10 \Rightarrow x = -4.$$

Multiple choice — medium

105 Factor $x^2 + 12x + 35$.

a. $(x + 9)(x + 5)$

b. $(x + 7)(x + 5)$ ✓

c. $(x + 7)(x - 3)$

d. $(x - 7)(x - 5)$

$(x + 7)(x + 5)$.

106 Factor $x^2 - 13x + 36$.

a. $(x - 2)(x - 9)$

b. $(x - 4)(x + 11)$

c. $(x - 4)(x - 9)$ ✓

d. $(x + 4)(x + 9)$

$(x - 4)(x - 9)$.

107 Factor $x^2 - 2x - 15$.

a. $(x - 3)(x + 5)$

b. $(x + 5)(x - 5)$

c. $(x + 3)(x - 5)$ ✓

d. $(x + 3)(x + 7)$

$(x + 3)(x - 5)$.

108 Factor $x^2 + 10x + 21$.

a. $(x + 7)(x - 1)$

b. $(x - 7)(x - 3)$

c. $(x + 7)(x + 3)$ ✓

d. $(x + 9)(x + 3)$

$(x + 7)(x + 3)$.

109 Factor $x^2 - x - 72$.

a. $(x - 9)(x - 6)$

b. $(x + 9)(x - 8)$

c. $(x - 9)(x + 8)$ ✓

d. $(x - 7)(x + 8)$

$(x - 9)(x + 8)$.

110 Factor $x^2 - 4x - 32$.

a. $(x - 6)(x + 4)$

b. $(x - 8)(x + 4)$ ✓

c. $(x + 8)(x - 4)$

d. $(x - 8)(x - 2)$

$(x - 8)(x + 4)$.

111 Factor $x^2 + 5x + 6$.

a. $(x - 3)(x - 2)$

b. $(x + 3)(x)$

c. $(x + 5)(x + 2)$

d. $(x + 3)(x + 2)$ ✓

$(x + 3)(x + 2)$.

112 Factor $x^2 + 3x - 28$.

- a. $(x + 9)(x - 4)$
- b. $(x + 7)(x + 6)$
- c. $(x - 7)(x + 4)$
- d. $(x + 7)(x - 4)$ ✓

$$(x + 7)(x - 4).$$

113 Factor $x^2 - 8x + 15$.

- a. $(x - 3)(x - 5)$ ✓
- b. $(x + 3)(x + 5)$
- c. $(x - 3)(x + 7)$
- d. $(x - 1)(x - 5)$

$$(x - 3)(x - 5).$$

114 Factor $x^2 + 5x - 14$.

- a. $(x + 9)(x - 2)$
- b. $(x + 7)(x + 4)$
- c. $(x - 7)(x + 2)$
- d. $(x + 7)(x - 2)$ ✓

$$(x + 7)(x - 2).$$

115 Factor $x^2 + 4x - 32$.

- a. $(x + 8)(x - 4)$ ✓
- b. $(x + 10)(x - 4)$
- c. $(x - 8)(x + 4)$
- d. $(x + 8)(x + 6)$

$$(x + 8)(x - 4).$$

116 Factor $x^2 + 11x + 18$.

a. $(x + 4)(x + 9)$

b. $(x + 2)(x - 7)$

c. $(x - 2)(x - 9)$

d. $(x + 2)(x + 9)$ ✓

$(x + 2)(x + 9)$.

117 Factor $x^2 + 6x - 16$.

a. $(x - 8)(x + 2)$

b. $(x + 10)(x - 2)$

c. $(x + 8)(x - 2)$ ✓

d. $(x + 8)(x + 4)$

$(x + 8)(x - 2)$.

118 Factor $x^2 - 36$.

a. $(x - 6)(x - 4)$

b. $(x - 4)(x + 6)$

c. $(x + 6)(x - 6)$

d. $(x - 6)(x + 6)$ ✓

$(x - 6)(x + 6)$.

119 Factor $x^2 - 9$.

a. $(x + 3)(x - 3)$

b. $(x - 3)(x + 3)$ ✓

c. $(x - 1)(x + 3)$

d. $(x - 3)(x - 1)$

$(x - 3)(x + 3)$.

Multiple choice — hard

120 Use the quadratic formula to solve $x^2 + x - 9 = 0$, leaving the answer in exact form.

a. $\frac{1 \pm \sqrt{37}}{2}$

b. $\frac{-1 \pm \sqrt{41}}{2}$

c. $\frac{-1 \pm \sqrt{33}}{2}$

d. $\frac{-1 \pm \sqrt{37}}{2}$ ✓

$$a = 1, b = 1, c = -9. x = \frac{-(1) \pm \sqrt{1 - (-36)}}{2} = \frac{-1 \pm \sqrt{37}}{2}.$$

121 Use the quadratic formula to solve $2x^2 - 9x + 8 = 0$, leaving the answer in exact form.

a. $\frac{-9 \pm \sqrt{17}}{4}$

b. $\frac{9 \pm \sqrt{13}}{4}$

c. $\frac{9 \pm \sqrt{21}}{4}$

d. $\frac{9 \pm \sqrt{17}}{4}$ ✓

$$a = 2, b = -9, c = 8. x = \frac{-(-9) \pm \sqrt{81 - 64}}{4} = \frac{9 \pm \sqrt{17}}{4}.$$

122 Use the quadratic formula to solve $x^2 - 8x - 5 = 0$, leaving the answer in exact form.

- a. $\frac{-8 \pm \sqrt{84}}{2}$
- b. $\frac{8 \pm \sqrt{84}}{2}$ ✓
- c. $\frac{8 \pm \sqrt{88}}{2}$
- d. $\frac{8 \pm \sqrt{80}}{2}$

$$a = 1, b = -8, c = -5. x = \frac{-(-8) \pm \sqrt{64 - (-20)}}{2} = \frac{8 \pm \sqrt{84}}{2}.$$

123 Use the quadratic formula to solve $2x^2 + 5x - 8 = 0$, leaving the answer in exact form.

- a. $\frac{-5 \pm \sqrt{93}}{4}$
- b. $\frac{-5 \pm \sqrt{85}}{4}$
- c. $\frac{-5 \pm \sqrt{89}}{4}$ ✓
- d. $\frac{5 \pm \sqrt{89}}{4}$

$$a = 2, b = 5, c = -8. x = \frac{-(5) \pm \sqrt{25 - (-64)}}{4} = \frac{-5 \pm \sqrt{89}}{4}.$$

124 Use the quadratic formula to solve $x^2 + 8x + 3 = 0$, leaving the answer in exact form.

- a. $\frac{-8 \pm \sqrt{56}}{2}$
- b. $\frac{8 \pm \sqrt{52}}{2}$
- c. $\frac{-8 \pm \sqrt{52}}{2}$ ✓
- d. $\frac{-8 \pm \sqrt{48}}{2}$

$$a = 1, b = 8, c = 3. x = \frac{-(8) \pm \sqrt{64 - 12}}{2} = \frac{-8 \pm \sqrt{52}}{2}.$$

125 Find the discriminant of $x^2 - 3 = 0$ and state how many real roots it has.

- a. 12, one repeated real root
- b. 12, no real roots
- c. 12, two distinct real roots ✓
- d. -12, two distinct real roots

$$\Delta = (0)^2 - 4(1)(-3) = 0 - (-12) = 12. \text{ Since } \Delta > 0, \text{ so there are two distinct real roots.}$$

126 Find the discriminant of $2x^2 - 2x + 4 = 0$ and state how many real roots it has.

- a. **-28, no real roots ✓**
- b. -28, one repeated real root
- c. 28, no real roots
- d. -28, two distinct real roots

$$\Delta = (-2)^2 - 4(2)(4) = 4 - 32 = -28. \text{ Since } \Delta < 0, \text{ so there are no real roots.}$$

127 Find the discriminant of $2x^2 - 6x + 5 = 0$ and state how many real roots it has.

- a. -4, two distinct real roots
- b. -4, one repeated real root
- c. **-4, no real roots ✓**
- d. 4, no real roots

$$\Delta = (-6)^2 - 4(2)(5) = 36 - 40 = -4. \text{ Since } \Delta < 0, \text{ so there are no real roots.}$$

128 Find the discriminant of $3x^2 - 4x + 9 = 0$ and state how many real roots it has.

- a. 92, no real roots
- b. **-92, no real roots ✓**
- c. -92, two distinct real roots
- d. -92, one repeated real root

$$\Delta = (-4)^2 - 4(3)(9) = 16 - 108 = -92. \text{ Since } \Delta < 0, \text{ so there are no real roots.}$$

129 Find the discriminant of $2x^2 - 8x - 2 = 0$ and state how many real roots it has.

- a. 80, no real roots
- b. 80, two distinct real roots ✓
- c. 80, one repeated real root
- d. -80, two distinct real roots

$\Delta = (-8)^2 - 4(2)(-2) = 64 - (-16) = 80$. Since $\Delta > 0$, so there are two distinct real roots.

130 Solve $|3x + 6| = 18$.

- a. 4, -6
- b. 4, -8 ✓
- c. -8, 4
- d. 6, -8

$3x + 6 = 18 \Rightarrow x = 4$, or $3x + 6 = -18 \Rightarrow x = -8$.

131 Solve $|3x| = 3$.

- a. -1, 1
- b. 1, -1 ✓
- c. 1, 1
- d. 3, -1

$3x = 3 \Rightarrow x = 1$, or $3x = -3 \Rightarrow x = -1$.

132 Solve $|x + 9| = 11$.

- a. -20, 2
- b. 4, -20
- c. 2, -20 ✓
- d. 2, -18

$x + 9 = 11 \Rightarrow x = 2$, or $x + 9 = -11 \Rightarrow x = -20$.

133 Solve $|x - 9| = 11$.

a. $-2, 20$

b. $22, -2$

c. $20, 0$

d. $20, -2$ ✓

$$x - 9 = 11 \Rightarrow x = 20, \text{ or } x - 9 = -11 \Rightarrow x = -2.$$

134 Solve $|x - 6| = 17$.

a. $-11, 23$

b. $23, -11$ ✓

c. $23, -9$

d. $25, -11$

$$x - 6 = 17 \Rightarrow x = 23, \text{ or } x - 6 = -17 \Rightarrow x = -11.$$